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This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	07 July 2017	First Release	Chin Chee Yan

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Removal Procedure

1. Unload board from machine if any.
2. Go to X ray source tab and toggle “Turn off X-Ray”. Power off at Operator Console Panel

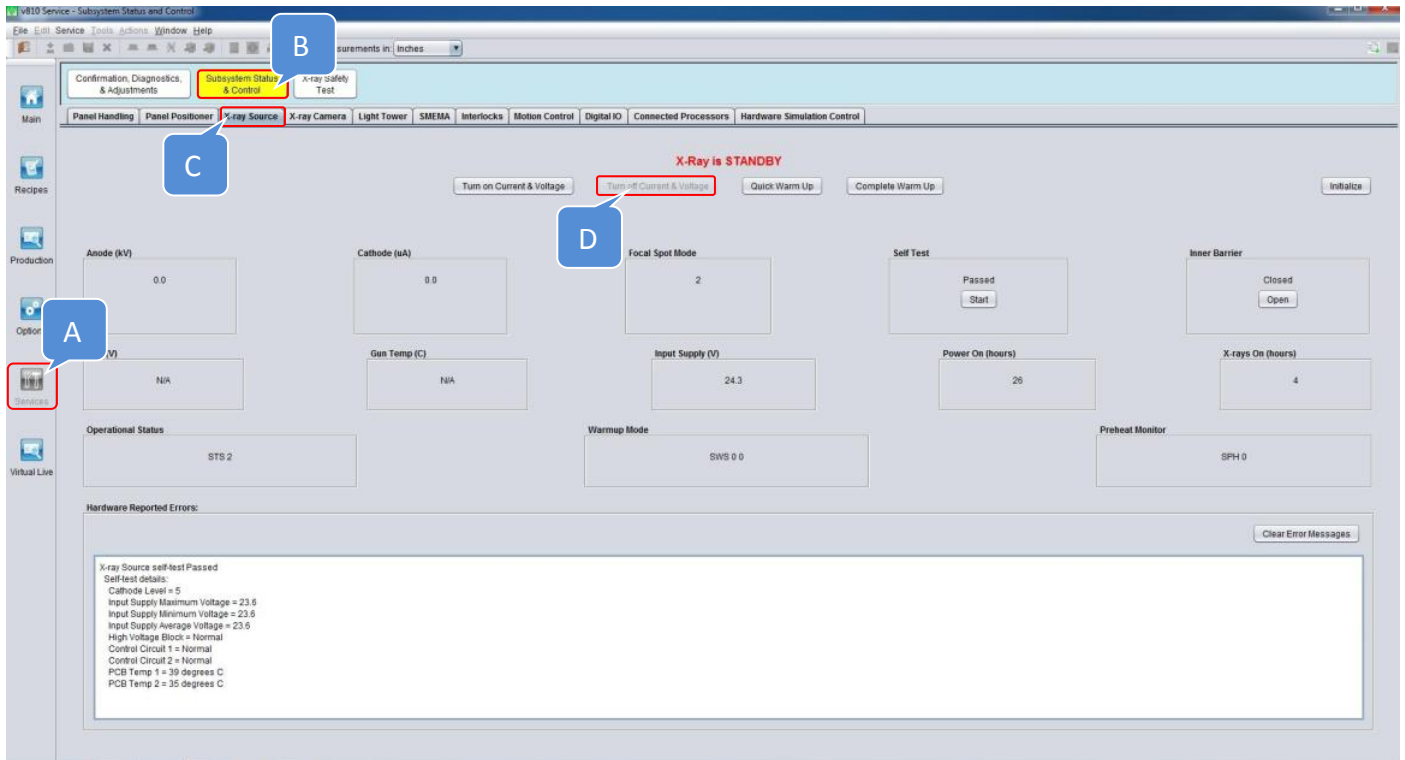


Figure 1 X Ray Source Shutdown

3. Power off SSAC

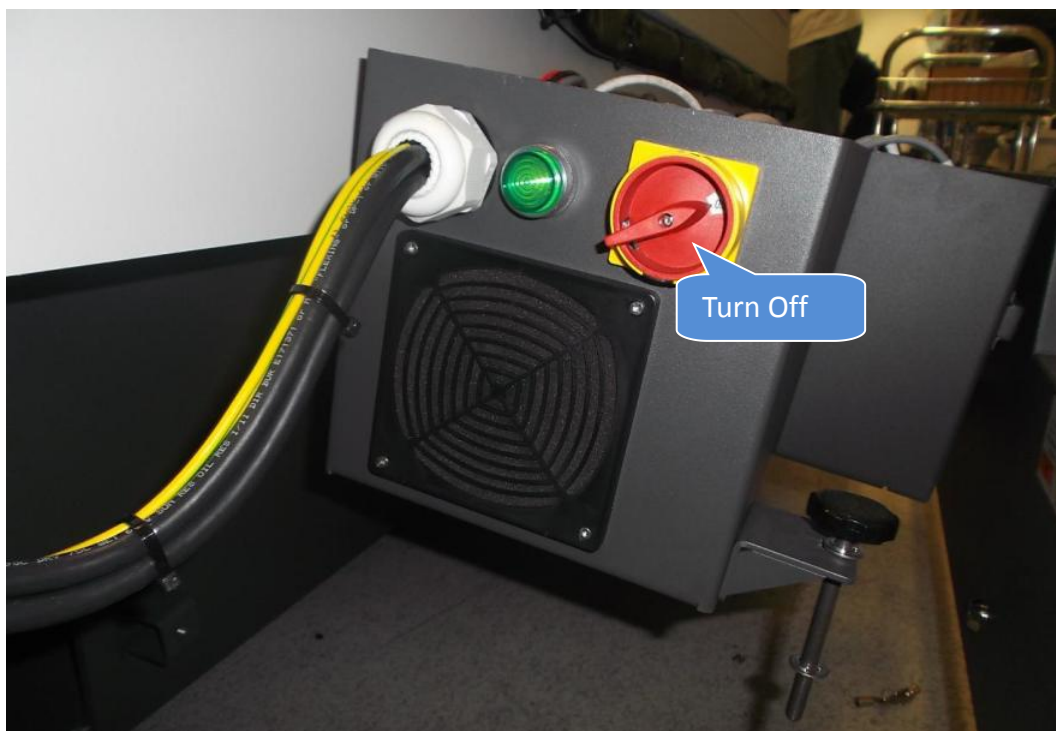


Figure 2 SSAC Power Off

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4. Gently loosen Both Anchors as below Figure 3



Figure 3 Anchors Release

5. Extend SSCA Support Rod which locate underneath of SSCA as below Figure 4



Figure 4 SSCA Support Rod Extend

I/O, MODULE, SYNQNET NETWORK ADAPTER Removal

6. Unlock Slice I/O, MODULE, SYNQNET NETWORK ADAPTER as below Figure 5

Remarks :Unlock side slice as well where mainly ease of removal process

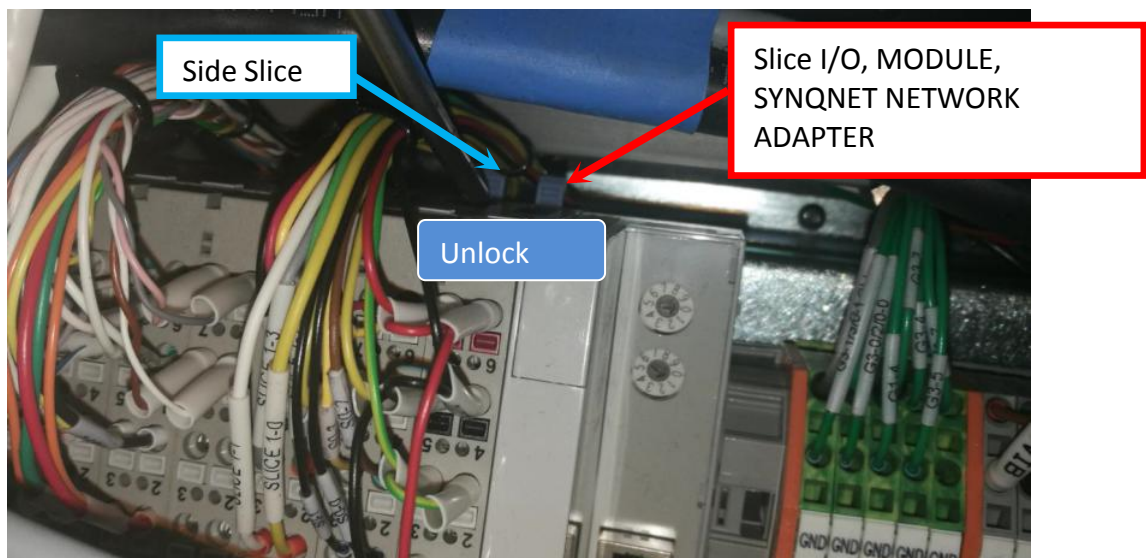


Figure 5 Slice (SYNQNET NETWORK ADAPTER) Unlock

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7. Probe Tiny Square Button with Flat Head Screwdriver to disconnect IO Cables.

Remarks : Perform Labeling at IO Cable upon removal. Applicable only for Cable Without Label

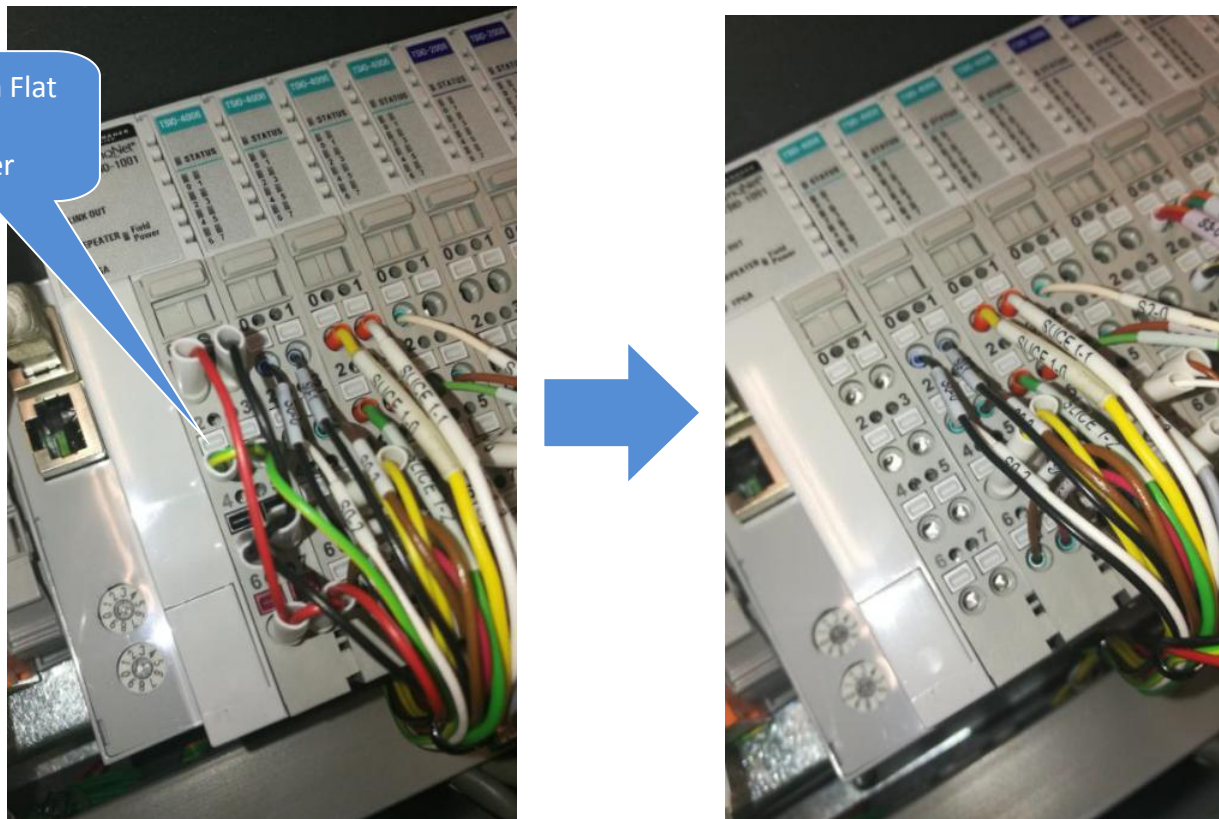


Figure 6 IO Cable Disconnect

8. Disconnect LAN cable as shown.

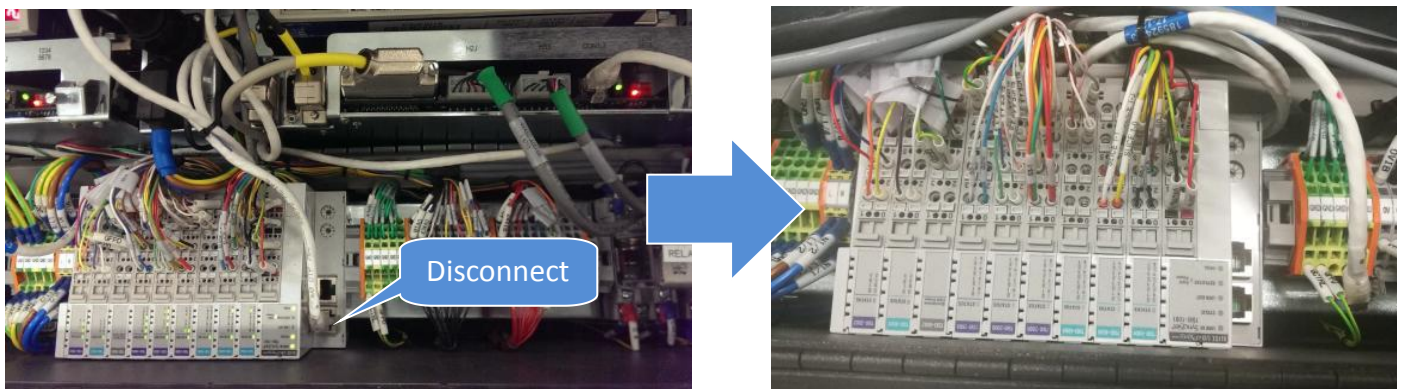


Figure 7 LAN cable Disconnect

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9. Access from Red Arrow Indicated. Gently pull Slice out from IO Module

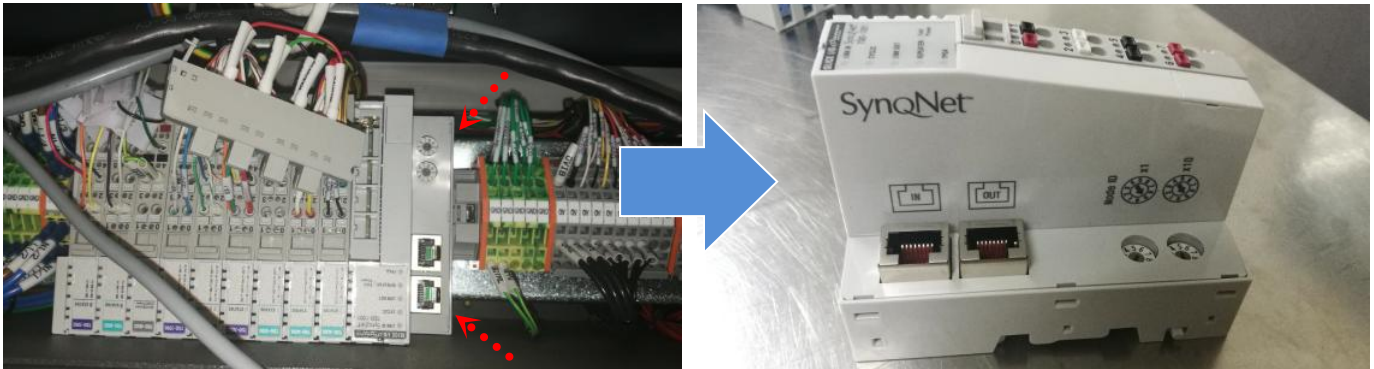


Figure 8 I/O, MODULE, SYNQNET NETWORK ADAPTER Removal

Selected Slice IO Removal

10. Unlock the lever as shown.

Remarks :Unlock side slices as well where mainly ease of removal process

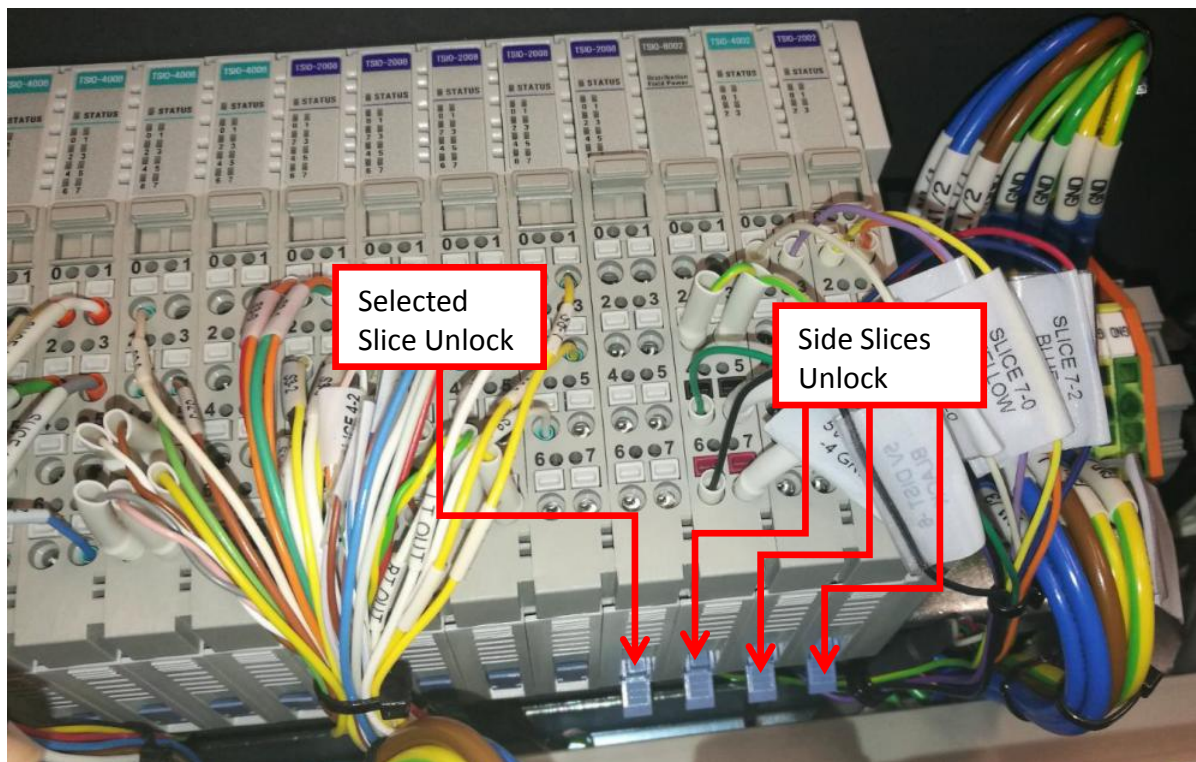


Figure 9 Selected Slice Unlock

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11. Gently probe the Tiny Square Button at Selected Slice IO. Disconnect all IO Cable from Selected Slice IO

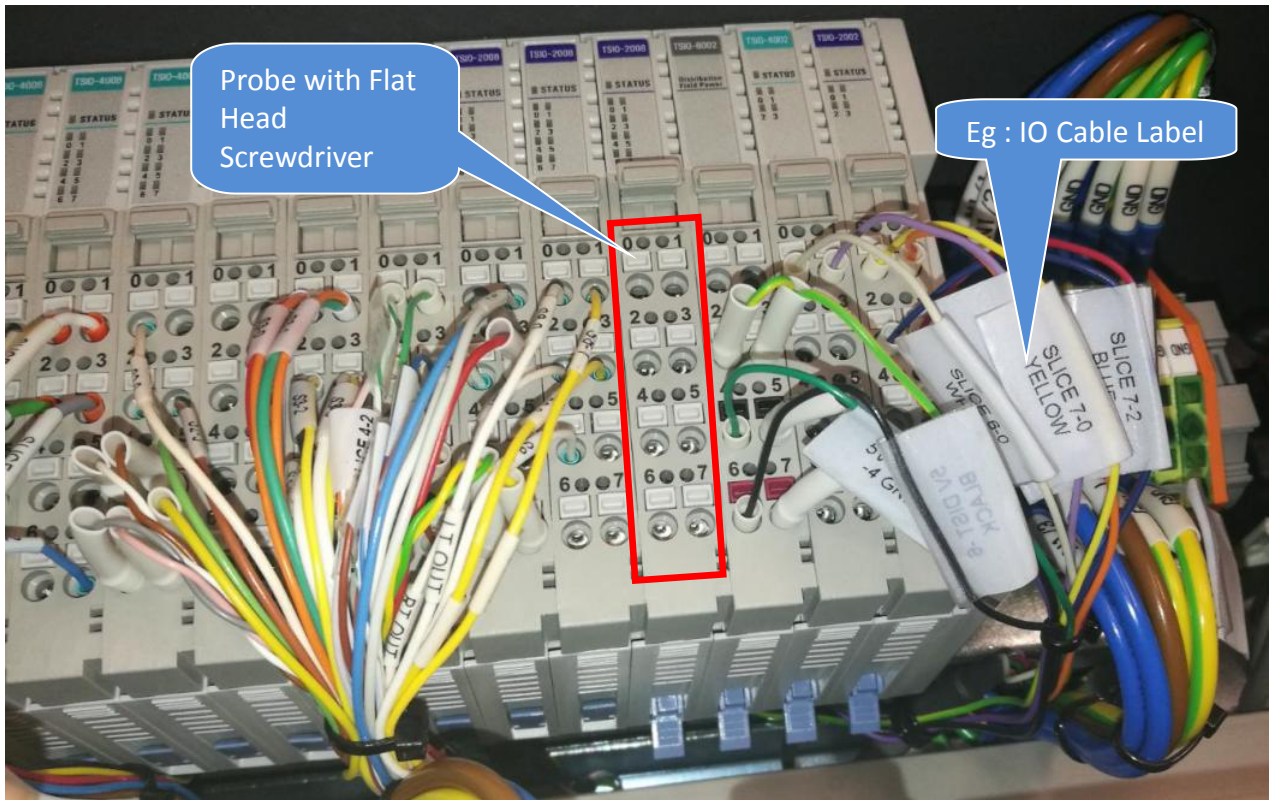


Figure 10 Selected IO Cable Removal

12. Remove the IO Module Side cover as shown.

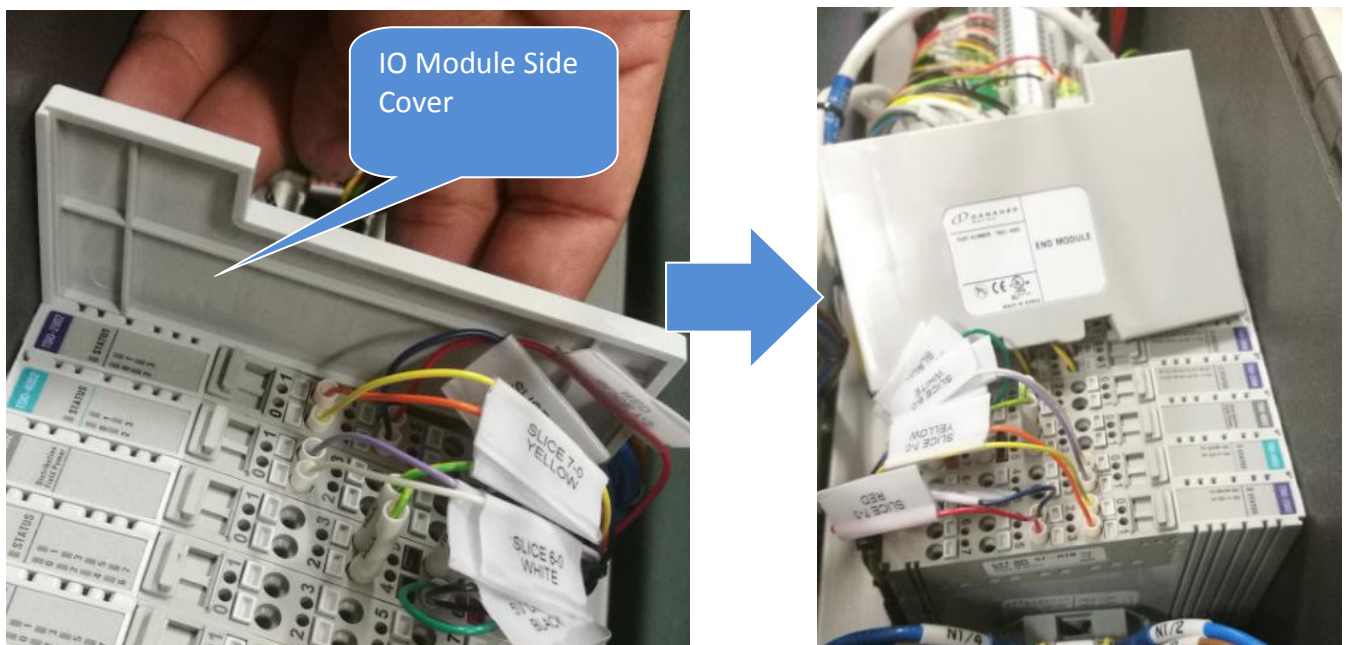


Figure 11 IO Module Side Cover Removal

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13. Access from the most right side. Remove All Slice IO Slice as below Figure 12

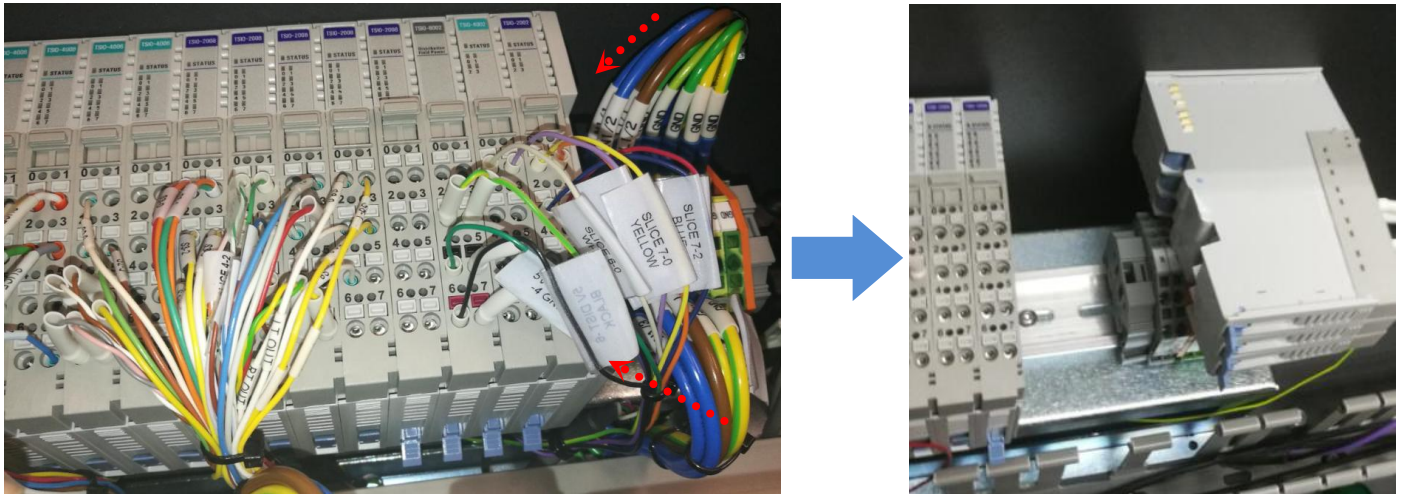


Figure 12 Side IO Slice Removal (Reference Only)

14. Remove the Selected Slice IO out from IO Module

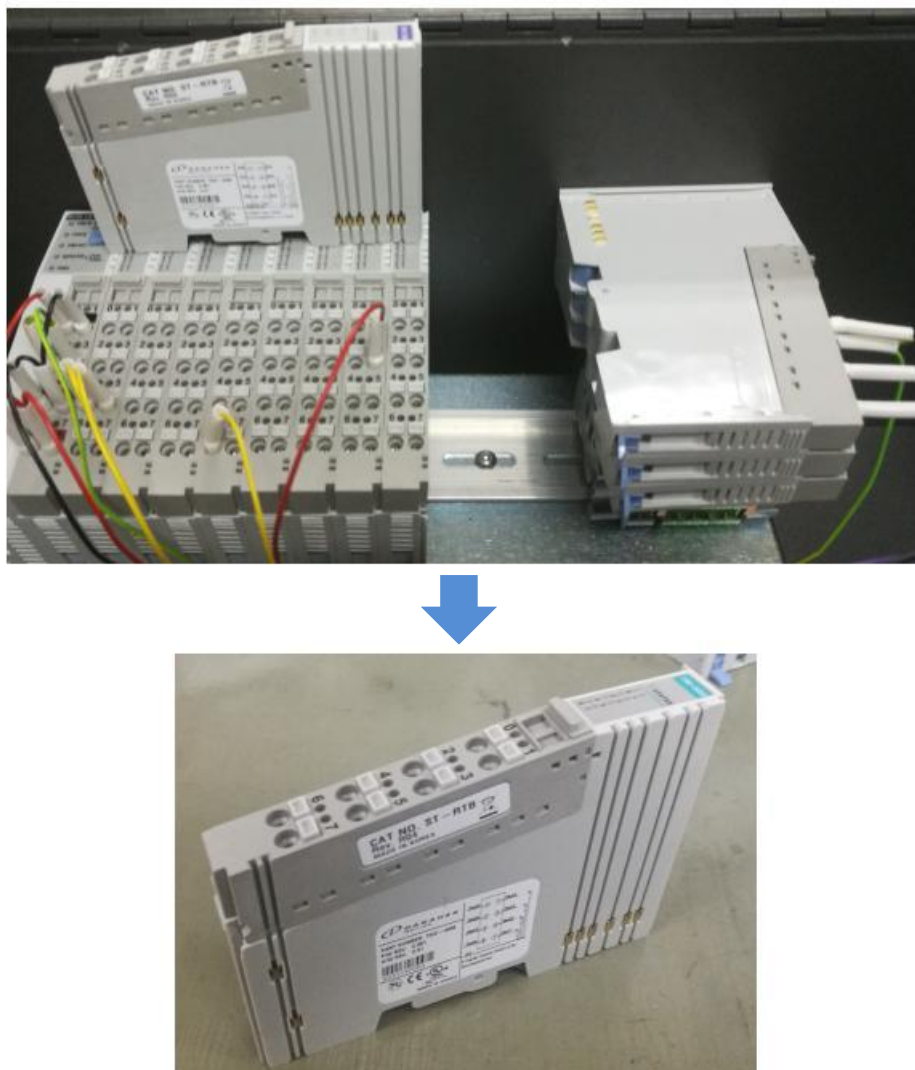


Figure 13 Selected Slice IO Removal

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Replacement Procedure

15. AXI V810 S2EX machine DIO Signal Pin Layout is illustrated as below.

S2EX																											
7-7	7-8	6-7	6-6	6	7	9-7	9-6	8-7	8-6	5-7	5-6	4-7	4-6	3-7	3-6	10-7	10-6	2-7	2-6	1-7	1-6	0-7	0-6	6	7		
DIO In - 31	DIO In - 30	DIO Out - 31	DIO Out - 30			DIO In - 47	DIO In - 46	DIO In - 39	DIO In - 38	DIO In - 23	DIO In - 22	DIO In - 15	DIO In - 14	DIO In - 7	DIO In - 6	DIO Out - 39	DIO Out - 38	DIO Out - 23	DIO Out - 22	DIO Out - 15	DIO Out - 14	DIO Out - 7	DIO Out - 6				
				5V	5V			Optical PP ErrorSignal	Optical PP Done Signal	SMEWA Conveyor Panel Available	SMEWA Conveyor Not Ready	Interlock Service Mode On	Interlock Powered On	L Op-PP Sensor Engaged	R Op-PP Sensor Engaged		Reset Optical PP Sensor	SHEWA Unlatched Panel Out	SHEWA Good Panel Out	Z Motor Enable		Inner Barrier Control Valve	Bottom Clamp Up	24V	24V		
7-6	7-4	6-5	6-4	4	5	9-5	9-4	8-5	8-4	5-5	5-4	4-5	4-4	3-5	3-4	10-5	10-4	2-5	2-4	1-5	1-4	0-5	0-4	4	5		
DIO In - 29	DIO In - 28	DIO Out - 29	DIO Out - 28			DIO In - 45	DIO In - 44	DIO In - 37	DIO In - 36	DIO In - 21	DIO In - 20	DIO In - 13	DIO In - 12	DIO In - 6	DIO In - 4	DIO Out - 37	DIO Out - 36	DIO Out - 21	DIO Out - 20	DIO Out - 13	DIO Out - 12	DIO Out - 5	DIO Out - 4				
				0V	0V	PH-Motor Alarm	PH-Axis Position Mag #2	Optical PP Installed	Clamp Open	Right Filter Height Sensor	Left Filter Height Sensor	Interlock Logic Bit #5	Interlock Logic Bit #4	R Panel Clear Sensor Clear	L Panel Clear Sensor Clear	PH-Motor Bit #2	PH-Motor Bit #1	SHEWA Tester Panel Available	SHEWA Tester Not Ready			Belt Motor L/R	Belt Motor On/Off	0V	0V		
7-3	7-2	6-3	6-2	2	3	9-3	9-2	8-3	8-2	5-3	5-2	4-3	4-2	3-3	3-2	10-3	10-2	2-3	2-2	1-3	1-2	0-3	0-2	2	3		
DIO In - 27	DIO In - 26	DIO Out - 27	DIO Out - 26			DIO In - 43	DIO In - 42	DIO In - 35	DIO In - 34	DIO In - 19	DIO In - 18	DIO In - 11	DIO In - 10	DIO In - 3	DIO In - 2	DIO Out - 35	DIO Out - 34	DIO Out - 19	DIO Out - 18	DIO Out - 11	DIO Out - 10	DIO Out - 3	DIO Out - 2				
Cameras Controller Phase Locked	Cameras Not Using Excessive Current			GND	GND	PH-Axis Position Mag #1	Z-Axis Position Mag #3	Right OSES (PP) (Retract)	Right OSES (PP) (Extend)	Z-Motor Output (+)	Z-Axis Position Loading	Interlock Logic Bit #3	Interlock Logic Bit #2	Right Outer Barrier Open	Left Outer Barrier Open	PH-Motor Bit #0	PH-Motor Start	Z-Motor A+	Z-Motor B+	Light Tower Buzzer	Light Tower Green	Right OSES (PP) Extended	Left OSES (PP) Extended	GND	GND		
7-1	7-0	6-1	6-0	1	0	9-1	9-0	8-1	8-0	5-1	5-0	4-1	4-0	3-1	3-0	10-1	10-0	2-1	2-0	1-1	1-0	0-1	0-0	1	0		
DIO In - 25	DIO In - 24	DIO Out - 25	DIO Out - 24			DIO In - 41	DIO In - 40	DIO In - 33	DIO In - 32	DIO In - 17	DIO In - 16	DIO In - 9	DIO In - 8	DIO In - 1	DIO In - 0	DIO Out - 33	DIO Out - 32	DIO Out - 17	DIO Out - 16	DIO Out - 9	DIO Out - 8	DIO Out - 1	DIO Out - 0				
Camera Control Power On	Camera Control Cables Connected	Cameras Control Spare	Cameras Synch Clock			Z-Axis Position Mag #2	Z-Axis Position Mag #1	Left OSES (PP) (Retract)	Left OSES (PP) (Extend)	Inner Barrier Open	Inner Barrier Closed	Interlock Logic Bit #1	Interlock Logic Bit #0	Right Outer Barrier Closed	Left Outer Barrier Closed	PH-Motor Home	PH-Motor Alarm Reset	Enable Optical PP Controller	Panel Clamps Open	Light Tower Yellow	Light Tower Red	Right Outer Barrier Control Valve	Left Outer Barrier Control Valve	0V	24V		
Slice 7 - Input	Slice 8 - Output	Field Power Distributor				Slice 9 - Input	Slice 8 - Input	Slice 6 - Input	Slice 4 - Input	Slice 3 - Input	Slice 10 - Output	Slice 2 - Output	Slice 1 - Output	Slice 0 - Output	MES Slice IO Controller												

Figure 14 DIO Signal Pin Block Layout

16. Table 1 is indicate AXI S2EX machine Slide IO Part Number (Red Column at Figure 14)

No	Slide	Part Number	Description
1	MES Slice IO Controller	2150-0020	I/O, MODULE, SYNQNET NETWORK ADAPTER (TSIO-1001)
2	Slice 0 - Output	2150-0021	I/O, MODULE, 8-CHANNEL +24V OUTPUT (TSIO-4006)
3	Slice 1 - Output	2150-0021	I/O, MODULE, 8-CHANNEL +24V OUTPUT (TSIO-4006)
4	Slice 2 - Output	2150-0021	I/O, MODULE, 8-CHANNEL +24V OUTPUT (TSIO-4006)
5	Slice 3 - Input	2150-0022	I/O, MODULE, 8-CHANNEL +24V INPUT (TSIO-2008)
6	Slice 4 - Input	2150-0022	I/O, MODULE, 8-CHANNEL +24V INPUT (TSIO-2008)
7	Slice 5 - Input	2150-0022	I/O, MODULE, 8-CHANNEL +24V INPUT (TSIO-2008)
8	Slice 8 - Input	2150-0022	I/O, MODULE, 8-CHANNEL +24V INPUT (TSIO-2008)
9	Slice 9 - Input	2150-0022	I/O, MODULE, 8-CHANNEL +24V INPUT (TSIO-2008)
10	Field Power Distributor	2140-0005	POWER SUPPLY, SILECE I/O DISTRIBUTOR (TSIO-8002)
11	Slice 6 - Output	2150-0023	I/O, MODULE, 4-CHANNEL +5V OUTPUT (TSIO-4002)
12	Slice 7 - Input	2150-0024	I/O, MODULE, 4-CHANNEL +5V INPUT (TSIO-2002)

Table 1 Slice I/O Part number

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I/O, MODULE, SYNQNET NETWORK ADAPTER Replacement

17. Prepare new I/O, MODULE, SYNQNET NETWORK ADAPTER (TSIO-1001) (P/N, 2150-0020) and unlock Lever as below Figure 15

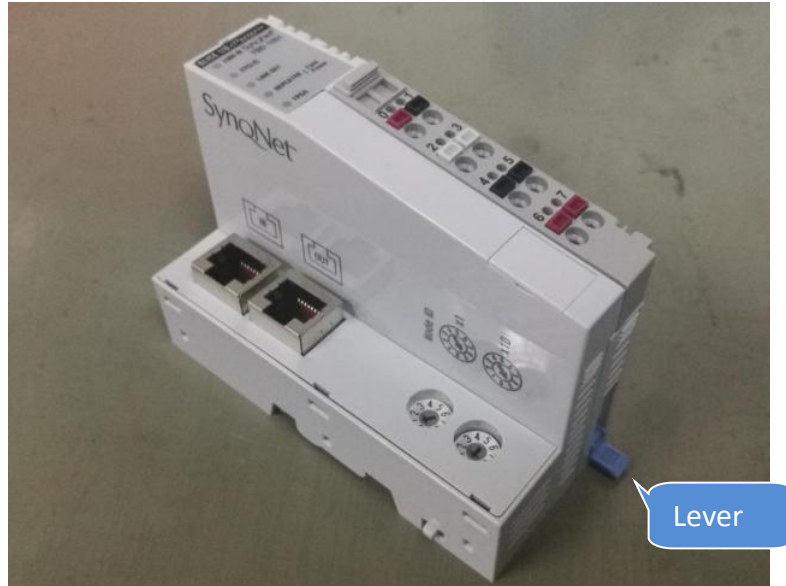


Figure 15 I/O, MODULE, SYNQNET NETWORK ADAPTER (TSIO-1001)

18. Replace new 2150-0020 by refer to below orientation. Slot on the track & gently push in & lock the lever accordingly.

Remarks : Lock the Side Slice IO as well if unlock previously

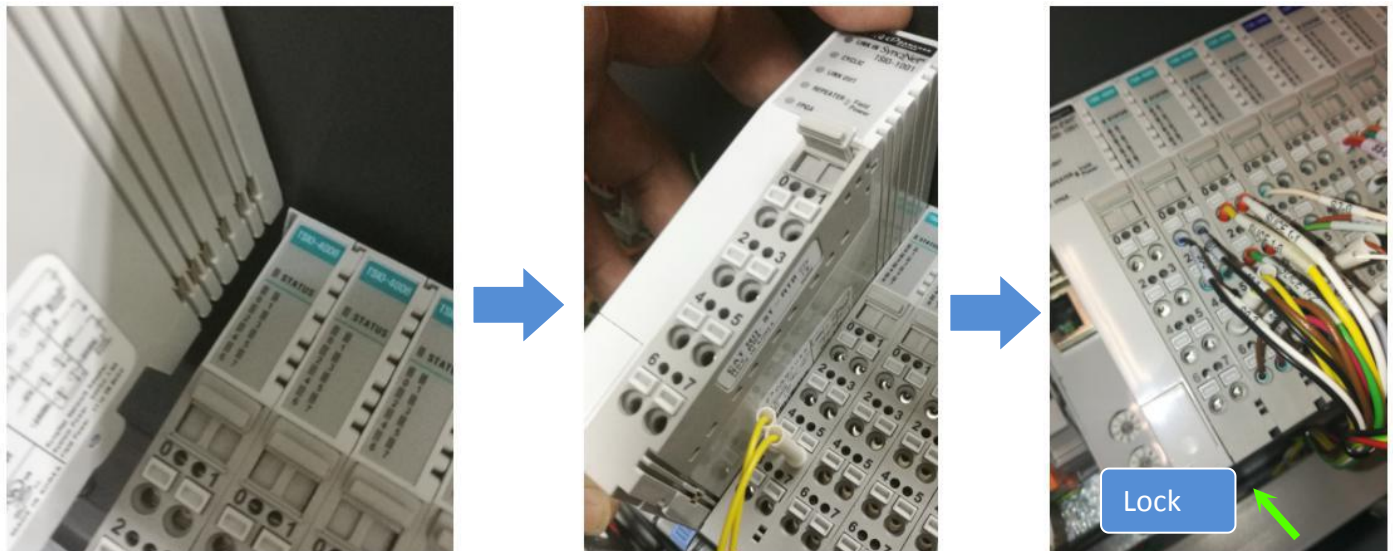


Figure 16

	Work Instruction	Doc. Revision	00
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19. While probe the Tiny Square Button, gently insert IO Cable respectively by refer to Figure 14

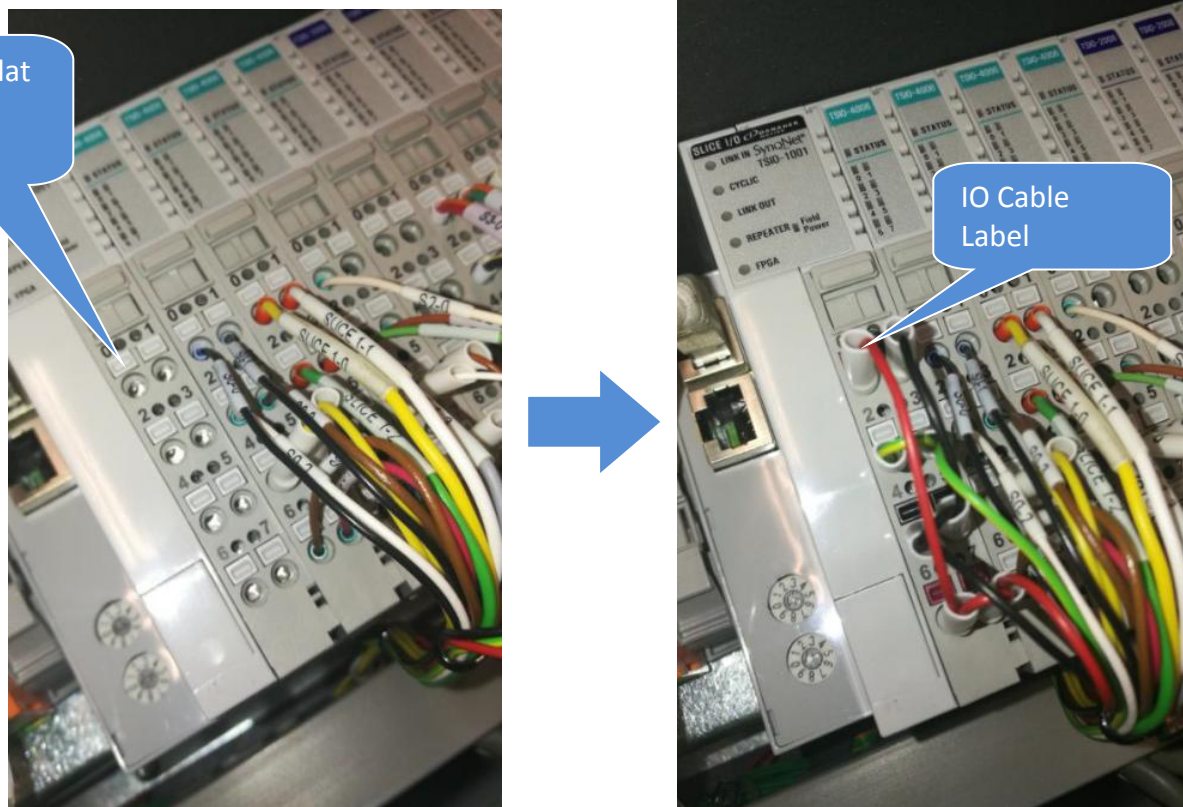


Figure 17 IO Cable Connection

20. Connect LAN cable to 2150-0020 "In" port as shown

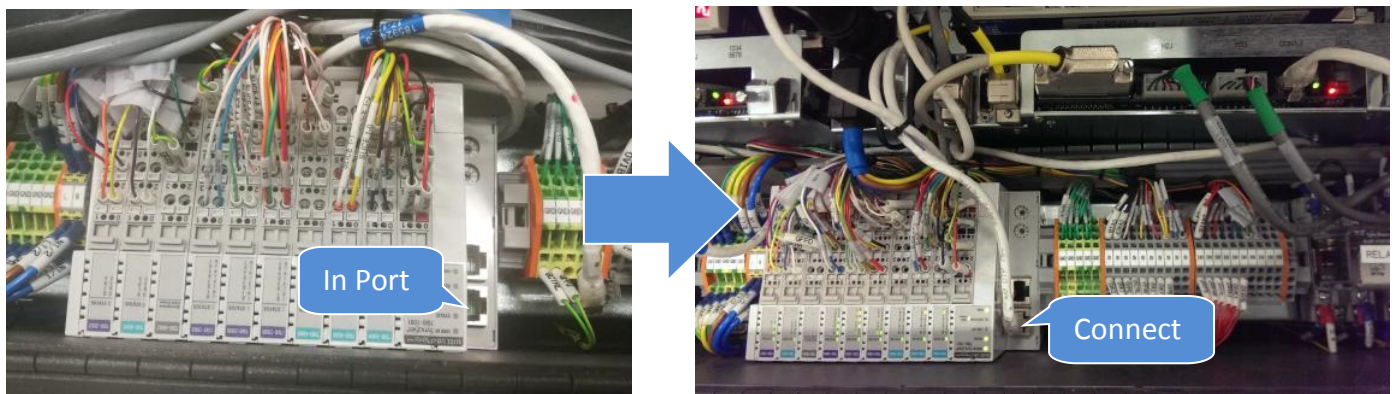


Figure 18 LAN Cable Connection

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Selected Slice IO Replacement

21. Prepare Selected Slice IO by refer to Table 1.Unlock the lever.



Figure 19 Selected Slice IO (Reference Picture Only)

22. Replace Selected Slice IO by refer to below orientation.Visual check pins where require firmly mating with Side Slice IO. Gently push in & lock the lever.

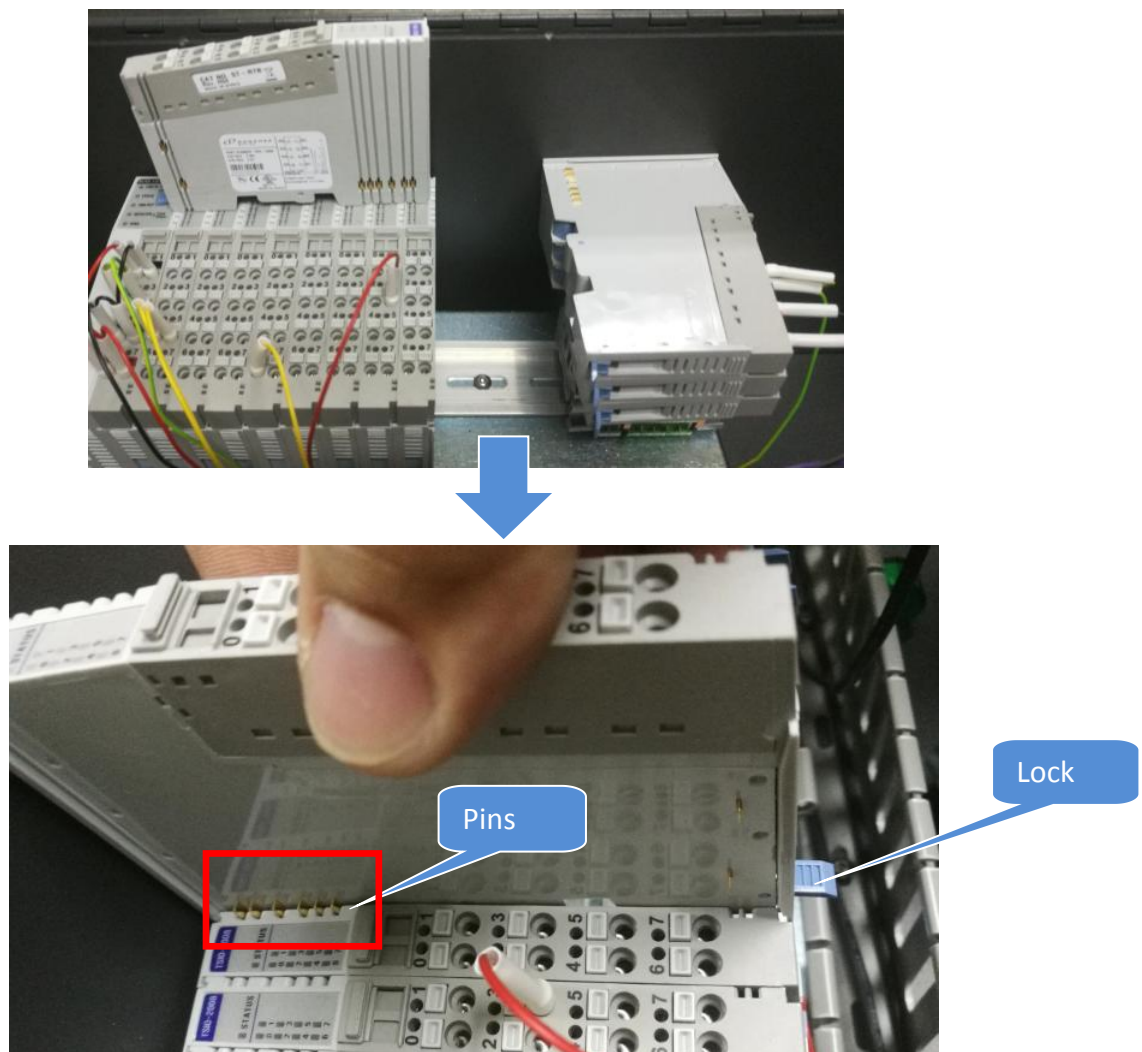


Figure 20

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23. Assemble all Side Slice IO respectively by refer to below orientation. Visual check pins where require firmly mating with Side Slice IO. Gently push in & lock the lever.

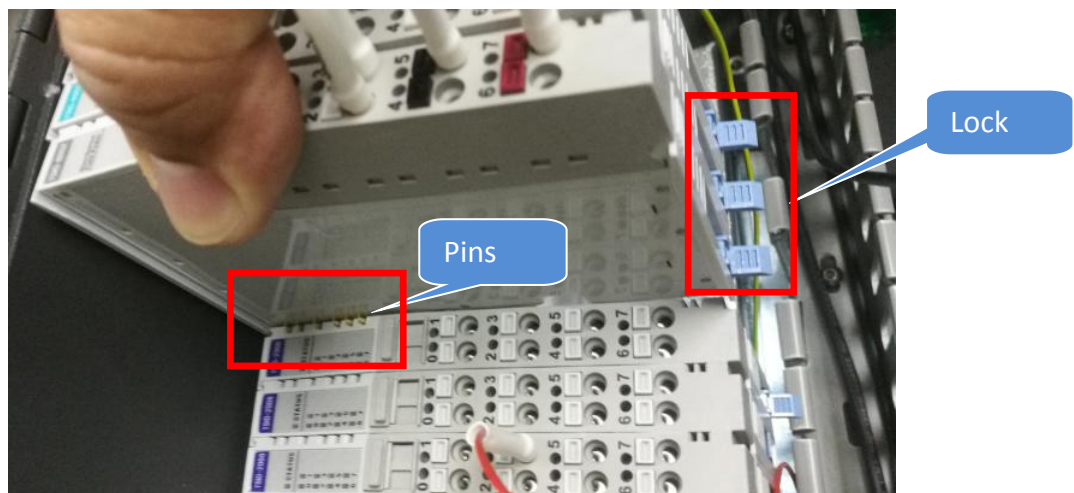


Figure 21 Side Slice IO Assembly

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24. Assemble IO Module Side Cover as shown. #Caution on cable snapping upon Side Cover Assembly

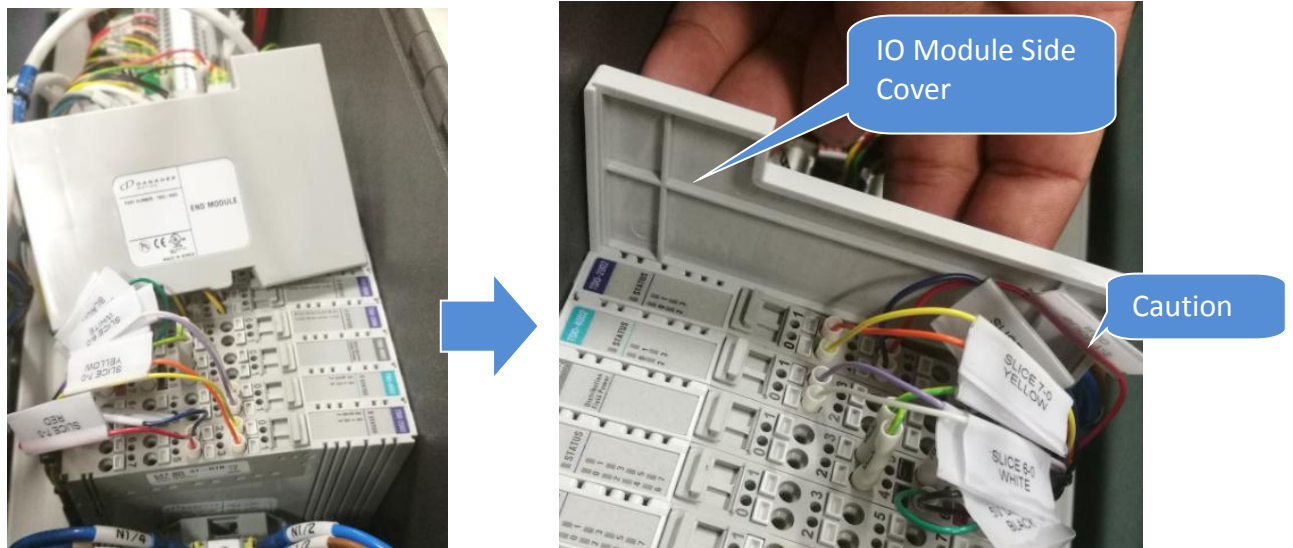


Figure 22 IO Module Side Cover Assembly

25. While probe the Tiny Square Button, gently insert IO Cable respectively by refer to Figure 23

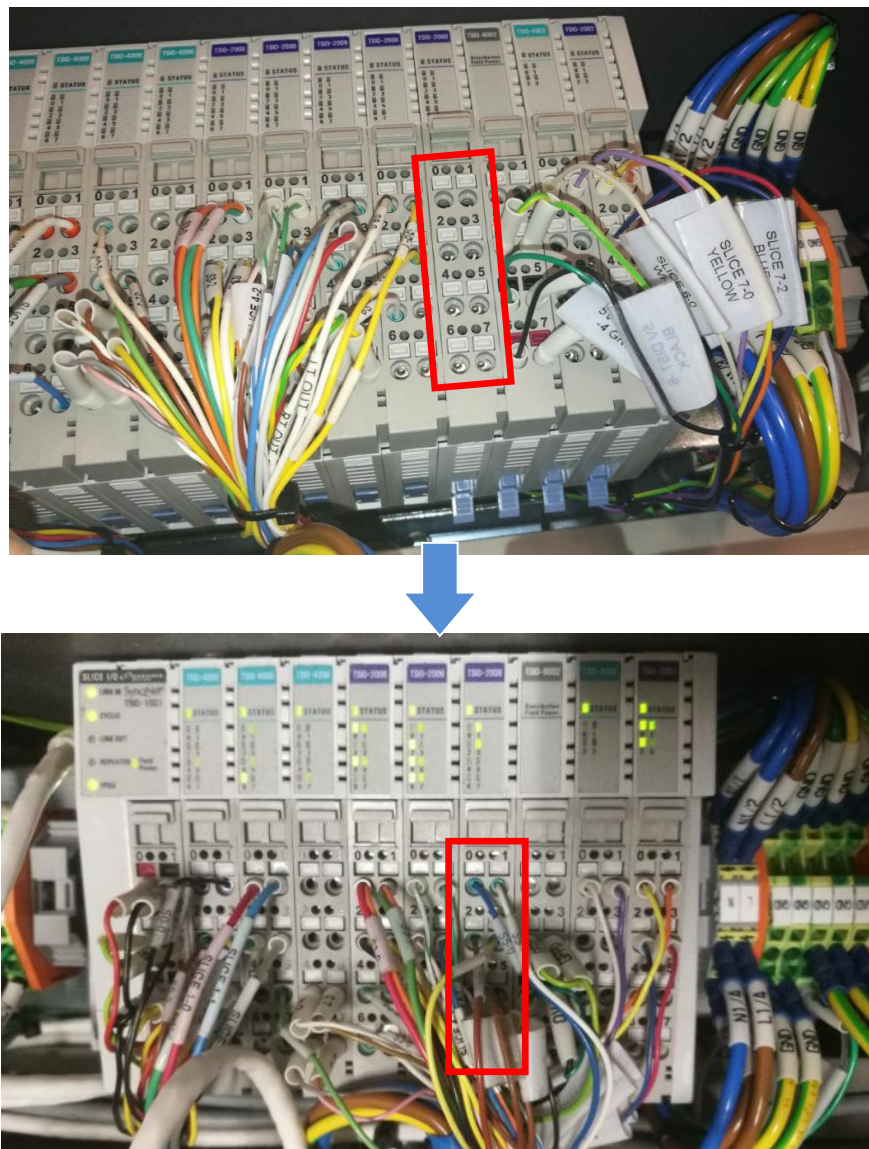


Figure 23

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Verification Procedure

26. Use Multi-meter to **probe the 0vDC and 24vDC Terminal block**, Ohm reading should approximately **130 Ohm and above**

If Ohm Reading Close to Zero , Short Circuit potentially occur . Do Not Power on SSCA. Re-screen all cable connection by refer to Step 18 - 24

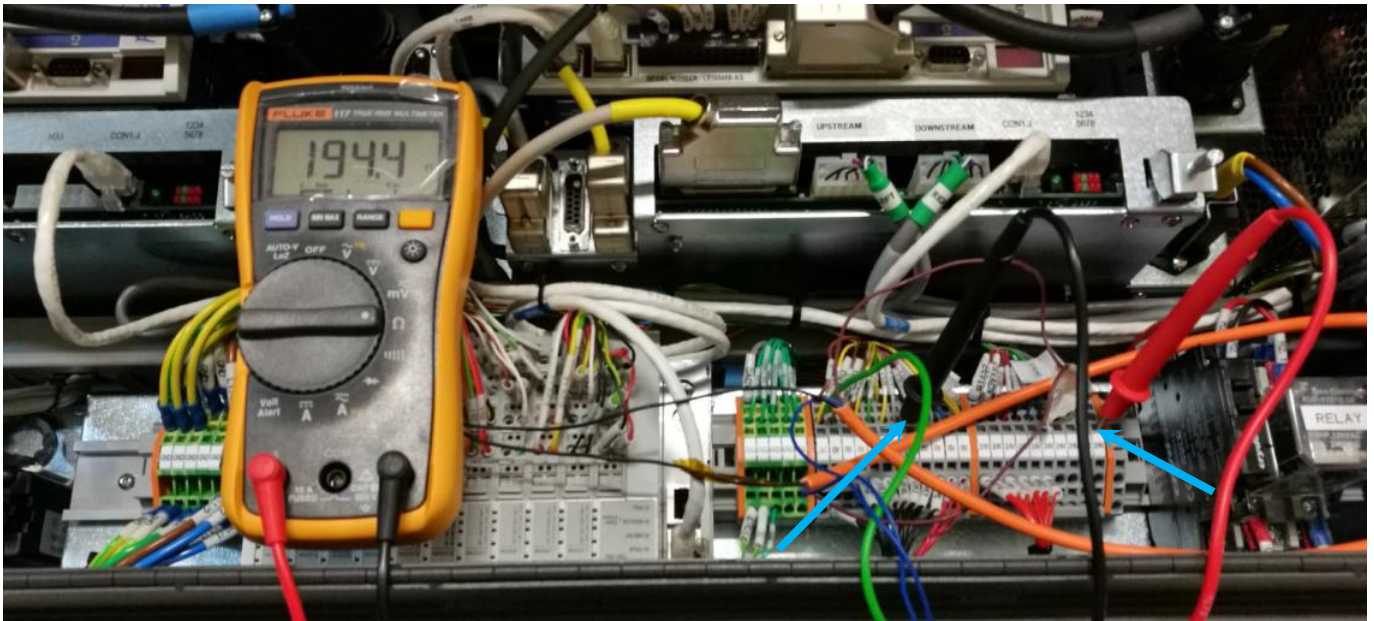


Figure 24 Terminal Block Verification

- 27. Open SSCA
- 28. Start-up system as usual
- 29. Run CDnA